

Faithful and Verifiable Reasoning in Language Models

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1. Research Problem and Central Idea

A language model can give the right answer for the wrong reason [1]. Consider a report placing a sensor failure at 10:02 and a power outage at 10:05. A model attributes the failure to the outage, contradicting the stated order. If the outage time is corrected to 9:58, the ordering objection disappears, but timing alone still does not establish the cause. A dependable reasoning system must distinguish these cases and explain what the evidence supports.

We ask: *How can a reasoning system check its conclusions and revise them when evidence changes?* Three capabilities will work together: make the argument explicit, choose which uncertain premises to check, and repair arguments whose support fails. A language model will propose interpretations and reasoning steps. A separate checker will assess the argument, from which the system will derive its accepted answer and explanation.

Our hypothesis is that explicit evidence dependencies help target checking and reduce the cost of revision. We will test whether this yields fewer unsupported answers at matched cost and answer coverage—the fraction of questions answered. Natural-language evidence is a demanding test: one misread statement can affect several otherwise valid inferences.

Foundation in the PI’s work. Prior methods for structured reasoning and logical interpretation of explanations inform the representation [3, 5]. MEQA supplies event-centered questions and explanations [6]. FG-PRM identifies errors within mathematical reasoning steps [7]; we will extend that supervision to interpreting evidence and repairing arguments. Work on controlling reasoning length informs the study of computational cost [8].

Research contribution. Symbolic solvers support explicit derivations, truth-maintenance methods track assumptions, and selective generation can control error by withholding answers [2, 4, 9, 10]. We will investigate how uncertain source interpretations affect these foundations. Sharing a check across arguments can save computation, but a misread shared premise can also spread error. We will study when checks can be reused, when alternatives require further validation, and which evidence changes require reconsidering a conclusion.

2. Research Aims and Technical Approach

Aim 1: Make the evidence and reasoning steps explicit

We will represent an argument as linked source statements, assumptions, and derived conclusions. Each premise will retain its source passage, relevant entities, and temporal conditions; each inference will identify the rule it applies. In the sensor example, the two event times and the requirement that a cause precede its effect would be recorded separately. This lets a checker locate the conflict instead of merely assigning the explanation a low score.

The initial scope will be finite arguments involving entity relations, event order, and conditional rules from a fixed library. A language model will propose a bounded set of interpretations and candidate arguments. Rule checks will assess each inference; evidence checks will compare its premises with source passages in context. An answer will be accepted only when a derivation passes both checks and retained plausible interpretations do not conflict about that answer. Otherwise, further checking or abstention is required. Unverified assumptions will remain explicit; conditional possibilities will be distinguished from supported answers.

We will study which information the representation must preserve for these checks to work. Controlled cases with known derivations will isolate mistakes such as confusing an event time with a report time or joining two different entities. The formal objective is to establish sufficient conditions for an accepted derivation to be valid within the restricted rule system. Any such guarantee will be conditional on the recorded premises. Evidence checks remain fallible, so we will measure interpretation errors and omitted alternatives separately. The final answer will be computed from the accepted derivation, with an explanation of the same steps. This defines faithfulness at the system level; the recorded argument need not describe every internal computation of the language model.

Aim 2: Decide which uncertain premises to check first

A system with limited time cannot repeatedly inspect every passage or explore every possible interpretation. If the timestamp in the sensor example is ambiguous, checking whether 10:05 refers to the outage or to the report could resolve it. If several arguments rely on that timestamp, the result of one check can be reused. This aim asks when checking a shared premise is more useful than examining another reasoning path.

We will begin with small argument graphs without cycles. Inexpensive rule checks will be applied to every candidate derivation. More costly checks will use relation extraction, textual entailment, and comparison of source passages to assess uncertain premises. Each check will have a measured cost and return supported, contradicted, or unresolved. We will develop policies that choose the next check using its estimated error-detection value, the conclusions it affects, and its cost. A starting comparison will be checking the most uncertain premise first versus considering its shared dependencies and alternative support. Learned scores will guide this choice; they will not count as proofs. The objective is to answer more questions at a specified error level within a budget, reporting which missing checks prevent a decision.

The formal analysis will start with chains and small graphs with known check costs and specified models of check outcomes. Exact solutions will provide reference policies for these restricted settings. We will study how shared premises and alternative derivations affect reliability and cost, seeking conditions under which efficient policies approach the reference performance. Experiments on larger graphs will test the limits of those conditions.

Reliability will be assessed for the complete acceptance procedure. Independent, human-validated cases will include shared-source errors and omissions of valid interpretations. Using a policy fixed before validation and cases drawn from the same distribution, we will apply finite-sample error bounds to determine how much validation data is needed to support a claim about accepted-answer reliability. New document sources will test transfer beyond those assumptions. If checking is too costly or uncertainty remains high, the system will withhold a conclusion; evaluation will measure the resulting loss of coverage.

Aim 3: Repair arguments when their support changes

A corrected premise should trigger reconsideration of the conclusions that use it. When the outage time changes from 10:05 to 9:58, the system should withdraw the temporal objection to the causal explanation. It should still require additional evidence before accepting that explanation. We will use recorded dependencies to identify affected arguments, preserve conclusions with other valid support, and explain what changed.

Classical truth maintenance provides the dependency-tracking foundation [2]. We will study its extension to premises whose source interpretation is uncertain. A repair may replace a misread premise, find another supported derivation, or withdraw the conclusion. The checking policy from Aim 2 will help choose among these actions. Every repair must pass the evidence and rule checks again; neither invented premises nor weakened rules will count as successful repairs.

We will train models to propose repairs from feedback on the failed step and its dependent conclusions. Training examples will pair verified arguments with controlled errors, including reversed event order, confused entity identities, and missing conditions. This adapts the structured-example generation and process supervision underlying the PI’s prior research. The central experiment will test whether this targeted feedback improves repair on unfamiliar combinations of rules compared with a generic request to try again.

For the initial finite graphs, we will analyze when revisiting only affected dependencies preserves the conclusions obtained by complete recomputation under the same premises and rules. A changed rule or a newly discovered dependency may require broader revision. Caps on repair attempts and per-attempt computation will bound the work performed. Reaching a cap will return an unresolved result, not a claim that repair succeeded. We will measure both the computation saved and errors introduced by revision.

3. Evaluation and Measures of Success

Evaluation will connect controlled reasoning problems with document-based questions. We will generate cases with known valid derivations, varying reasoning depth, shared premises, alternative support, and missing or conflicting evidence. Text versions will be checked independently against the underlying facts. MEQA will provide an existing document-based testbed, extended with human-reviewed ambiguities and evidence corrections. Cases that cannot be represented by the rule library will count as unresolved

rather than being removed from evaluation. Held-out rule combinations and document sources will test generalization.

Comparisons will include direct language-model reasoning, FG-PRM step scoring, solver-based reasoning, selective generation, and truth maintenance with exhaustive or uniform checking. Repair will also be compared with complete recomputation. We will match the base model, evidence access, and total computational budget, including generation, checking, and repair. We will compare error at matched answer coverage and report reliability-cost curves. Ablations will isolate the effects of explicit dependencies, check selection, and training with error-specific feedback.

The main outcomes will be unsupported-answer rate, answer accuracy and coverage, evidence-interpretation accuracy, inference validity, and repair success and cost. An accepted answer is unsupported if its disclosed derivation contains an invalid inference or a premise not justified by the supplied evidence. Conditional possibilities will not count toward accepted-answer coverage. We will also change necessary premises and test whether the accepted answer changes as required, scoring cases with alternative valid support separately. Human review will assess whether explanations accurately describe the accepted derivation. Success will mean fewer unsupported answers at comparable coverage and cost, and cheaper correct updates when a small part of the evidence changes. We will report uncertainty intervals and cases where restricted formal results fail to predict behavior on documents.

4. Relevance, Team, and Development Plan

The project addresses the Computational Cognition and Machine Intelligence portfolio’s interests in interpretable foundations, formal verification, uncertainty, and reasoning that completes within resource limits. Its Air Force and Space Force relevance lies in drawing defensible conclusions from incomplete or changing reports. The expected scientific outcome is a clearer account of when explicit evidence dependencies make reasoning more reliable and revision more efficient.

PI Xinya Du will lead the work at UT Dallas with graduate and undergraduate research students. The estimated three-year budget is \$450,000, including institutional indirect costs, for personnel, computation, annotation, and dissemination. No external collaborators or subawards are planned.

Year 1 will establish the representation, rule library, conditional-validity analysis, and controlled evaluation cases. **Year 2** will develop check-selection policies and analyze reliability and cost in the restricted settings. **Year 3** will develop and evaluate repair, integrate the three aims, and test transfer to new document sources. Deliverables will include reproducible implementations, annotated evaluation data, checker specifications, research publications, and conference presentations.

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